page N 1133 Exercise 35

$x = u + v \quad y = 3u^2 \quad z = u - v \quad (2,3,0)$

$r(u,v) = (u+v)i + 3u^2 j + (u-v)k$

$r_u = i + 6u j + k$

$r_v = i - k$

$r_u \times r_v = -6u\, i + 2j - 6u k$

$(2,3,0) \Rightarrow u = 1, v = 1$

$-6i + 2j - 6k$

tangent plane $\Rightarrow -6x + 2y - 6z = -6$

$3x - y + 3z = 3$

page N 1133, exercise 45

$x = u^2 \quad y = uv \quad z = \frac{1}{2}v^2 \quad 0 \leq u \leq 1$

$v \in [0, 2]$

$r_u = \langle 2u, v, 0 \rangle$

$r_v = \langle 0, u, v \rangle$

$r_u \times r_v = \langle v^2, -2uv, 2u^2 \rangle$

$A(S) = \iint_D |r_u \times r_v|\, dA = \int_0^1 \int_0^2$

$\sqrt{v^4 + 4u^2v^2 + 4u^4}\, du\, dv =$

$$= \int_0^1 \int_0^2 \sqrt{(v^2+2u^2)^2}\, dv\, du = \int_0^1 \int_0^2 (v^2+2u^2)\, dv\, du = \int_0^1 \left(\frac{1}{3}v^3 + 2u^2 v\right)\Big|_{v=0}^{v=2} du =$$

$$= \int_0^1 \left(\frac{8}{3} + 4u^2\right) du = \left(\frac{8}{3}u + \frac{4}{3}u^3\right)\Big|_0^1 = 4$$

page 1144 exercise n°5

$$\iint_S (x+y+z)\, dS \qquad \begin{aligned} x &= u+v \\ y &= u-v \\ z &= 1+2u+v \\ 0 &\le u \le 2 \\ 0 &\le v \le 1 \end{aligned}$$

$$r(u,v) = (u+v)\,i + (u-v)\,j + (1+2u+v)\,k$$

$$r_u \times r_v = (i + j + 2k) \times (i - j + k) = 3i + j - 2k$$

$$|r_u \times r_v| = \sqrt{3^2 + 1^2 + (-2)^2} = \sqrt{14}$$

$$\iint_S (x+y+z)\, dS = \iint_D (u+v+u-v+1+2u+v)\, |r_u \times r_v|\, dA = \int_0^1 \int_0^2 (4u+v+1) \cdot \sqrt{14}\, du\, dv = \sqrt{14} \int_0^1 (2u^2 + uv + u)\Big|_{u=0}^{u=2} dv$$

$$= \sqrt{14} \int_0^1 (2v + 10)\, dv = \sqrt{14}\,(v^2 + 10v)\Big|_0^1 =$$

$$= 11\sqrt{14}$$

page 1145 exercise 23

$F(x,y,z) = xy\,i + yz\,j + zx\,k$

$z = g(x,y) = 4 - x^2 - y^2$

$z = 4 - x^2 - y^2$

$0 \le x \le 1$

$0 \le y \le 1$

D is square $(0,1)\times(0,1)$

$\iint_S F\,dS = \iint_D (-xy(-2x) - yz(-2y) - zx)\,dA =$

$= \int_0^1 \int_0^1 (2x^2y + 2y^2(4 - x^2 - y^2) + x(4 - x^2 - y^2))\,dy\,dx = \int_0^1 \left(\frac{1}{3}x^2 + \frac{11}{3}x - x^3 + \frac{34}{15}\right)dy$

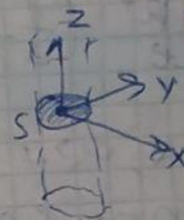
$= \frac{713}{180}$

Q 21

R - radius of vessel

a) $\iint \vec{F} \cdot d\vec{S} = ?$

$S: z-d$



$u \in [0;R]$

$v \in [0;2\pi]$

$$\vec{F} = \begin{pmatrix} 0 \\ 0 \\ \frac{\rho}{4\eta e}(R^2 - x^2 - y^2) \end{pmatrix}$$

$$\vec{r}(u,v) = \begin{pmatrix} x(u,v) = u\cos v \\ y(u,v) = u\sin v \\ z(u,v) = 0 \end{pmatrix} \Rightarrow x^2 + y^2 = u^2$$

$$0 \le u \le 1 \Rightarrow \vec{r}\,'_u = (\cos v, \sin v, 0)$$

$$\vec{r}\,'_v = (-u\sin v, u\cos v, 0)$$

$$u \times v = (0, 0, u\cos^2 u + u\sin^2 v) =$$

$$= (0, 0, u)$$

$$\iint_{S: z-d} F\, ds = \iint_{(u,v)} \frac{\rho}{4\eta e}\left(\frac{R^2 - x^2 - y^2}{u^2}\right) u\, du\, dv$$

$$\frac{\rho}{4\eta} \iint (R^2 - u^2)\, u\, dv\, du = \frac{\rho\pi}{2\eta e} \int_0^R (R^2 u - u^3)\, du$$

$$= \frac{\rho\pi R^4}{2\eta e}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{R^2 u^2}{2} - \frac{u^4}{4}$$

Q 21, B

$$\frac{P \Pi R^4}{2 \eta \ell}\left(\frac{1}{2}-\frac{1}{4}\right) \approx 0{,}000119$$

$\eta = 0{,}027$

$R = 0{,}008$ cm

$\ell = 2$ cm

$P = 4000$ dynes/cm^2